Sahaj Bhadja

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Summary

AI/ML engineer with 2+ years applying **Python, PyTorch, NLP, generative AI, data preprocessing, and model evaluation** to research and production workflows, with strong foundations in algorithms, APIs, and cloud delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built an AI education platform for 500+ users, reducing dashboard latency 71% through **Python services, structured data preprocessing, normalized schemas, feature-oriented queries, validation, and production performance analysis**.
- Built asynchronous FastAPI orchestration processing 1K+ concurrent model jobs at 3x throughput through **batch pipelines, failure isolation, experiment tracking, monitoring, model-service debugging, and reproducible production inference workflows**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **AI application integration, validation, and production releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, reducing turnaround 60% through **unstructured-data preprocessing, feature extraction, validation, automated testing, deployment, and model-pipeline error analysis**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Developed Decision Transformer pipelines processing 100K+ autonomous-driving trajectories, improving planning 18% through **PyTorch, NumPy, feature engineering, temporal data preprocessing, reproducible experimentation, benchmark analysis, and quantitative model evaluation**.
- Integrated HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% while **tuning deep-learning architectures, analyzing errors, visualizing predictions, and documenting research findings**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and cloud data pipelines using **Python, AWS Glue, S3, SQL, structured transformations, and scalable data integration**, improving response times 30% while reducing technical debt 20%.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **pipeline validation, regression coverage, reproducibility, and maintainable AI delivery**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React and Node.js workflows for resources, billing, permissions, and notifications, synchronizing 2M+ records through **reusable TypeScript components, REST APIs, WebSockets, authentication, and a shared operational database**.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **user-impact investigation, product analytics, root-cause debugging, and reliable weekly releases**.

Skills

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Predictive Analytics, Feature Engineering, Data Preprocessing, Exploratory Data Analysis, Model Evaluation, Hyperparameter Tuning
AI & NLP: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, LLMs, Generative AI, RAG, Prompt Engineering, Embeddings, FAISS, Pinecone, Natural Language Processing
Data & Programming: Python, Pandas, NumPy, SQL, PostgreSQL, Data Structures, Algorithms, Matplotlib, Seaborn, Jupyter, Structured Data, Unstructured Data
Applications: FastAPI, Flask, REST APIs, Model Serving, Batch Pipelines, Data Pipelines, React, TypeScript, Git, GitHub, Unit Testing, Integration Testing, Experiment Tracking
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Monitoring, Debugging, Technical Documentation, Agile, Model Performance Analysis

Projects

Customer Churn Intelligence Pipeline | *Python, Pandas, Scikit-learn, XGBoost, MLflow, FastAPI*

- Built an **end-to-end predictive analytics pipeline** performing exploratory analysis, missing-value treatment, feature engineering, class balancing, and model comparison, then serving calibrated churn probabilities through FastAPI with reproducible MLflow experiments.

Multimodal Support Copilot | *PyTorch, Transformers, RAG, FAISS, FastAPI, Docker*

- Built a **generative AI support assistant** preprocessing documents and images, retrieving evidence with embeddings and FAISS, evaluating grounded responses, tracing failure cases, and exposing containerized inference through tested REST APIs.

Big-Brain | *LLaMA 2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **retrieval-augmented AI knowledge product** indexing 50K+ documents with LLaMA 2, FAISS, Pinecone, PostgreSQL, and REST APIs to deliver grounded answers through a deployed GCP-backed application.